BUSINESS REPORT

Insurance Claim Prediction & Risk Management

Stage 4 — Dashboard and Business Report

58,592	3,748	6.40%	0.647
Policies	Observed claims	Observed claim rate	Final test ROC-AUC

1. Executive Summary

This Stage 4 report converts the insurance claim analysis and final machine-learning model into a management-oriented decision-support framework. The analysis covers 58,592 policies, of which 3,748 have an observed claim, giving an overall claim rate of approximately 6.40%.

The Tuned Decision Tree is the selected final model. On the held-out test set it achieved ROC-AUC 0.647, precision 0.089, recall 0.652 and F1-score 0.157. These are the authoritative final test metrics used in this report. The previously observed prediction-file figure of 0.731 recall is deliberately not reported as a final model metric because it has not been reconciled with the source evaluation output.

The business implication is clear: the model is more suitable for prioritising policies for underwriting review and risk-management attention than for making fully automated accept/reject decisions. Its low precision means that many flagged policies will not ultimately produce claims.

2. Problem Statement, Business Context & Success Metrics

Motor insurers need to identify policies with elevated likelihood of claims so that underwriting checks, risk monitoring and preventive interventions can be focused where they are most valuable. The analytical challenge is particularly important because claims represent a minority of policies; therefore, accuracy alone can be misleading.

Success metric	Business purpose
ROC-AUC	Measure how well the model ranks higher-risk observations above lower-risk observations.
Recall	Measure how many actual claims are captured by the model.
Precision	Measure how many model-flagged cases are actual claims.
F1-score	Balance precision and recall for the imbalanced claim class.
Operational usability	Convert model probability into review-priority risk tiers.

3. Data Summary & Preprocessing

Item	Summary
Dataset	58,592 rows × 49 columns
Target	Insurance claim indicator
Observed claims	3,748
Observed claim rate	6.40%

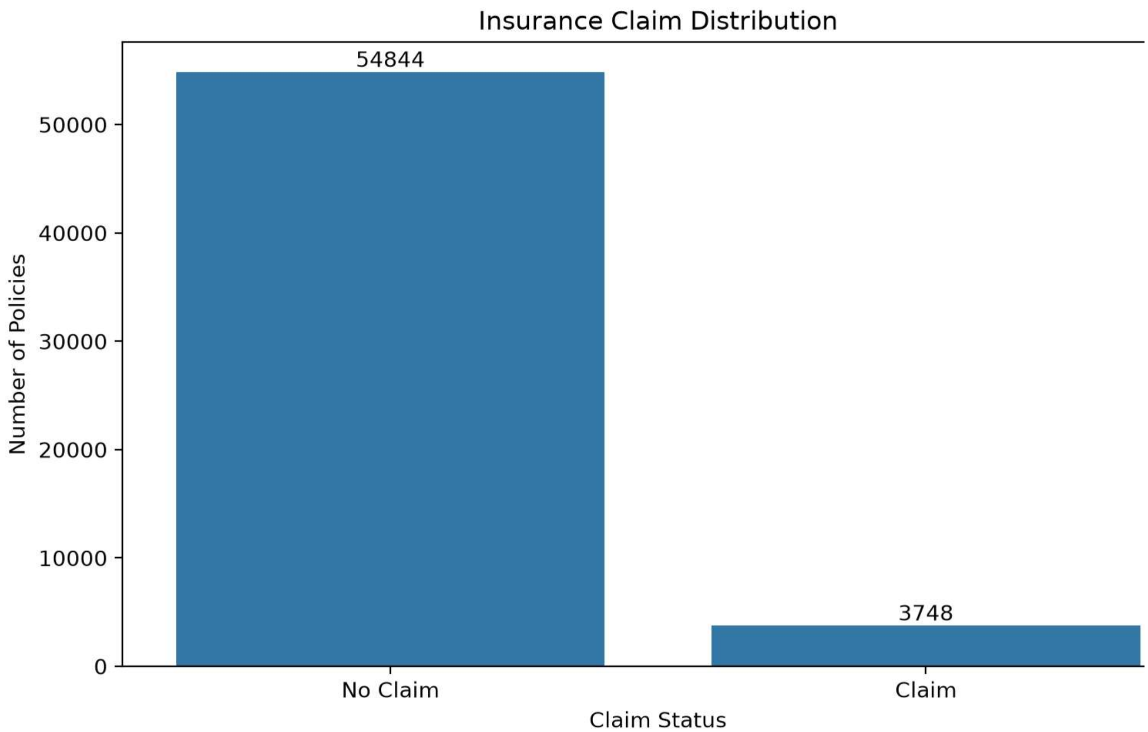
Numerical features	37
Categorical features	11

The modelling workflow included schema and missing-value checks, duplicate assessment, target-balance analysis, feature engineering, categorical/numerical preprocessing and feature-reduction diagnostics. The modelling pipeline was designed to prevent information leakage by fitting preprocessing using the training data rather than the held-out test data.

4. Key EDA & Feature Engineering Findings

4.1 Claim Distribution

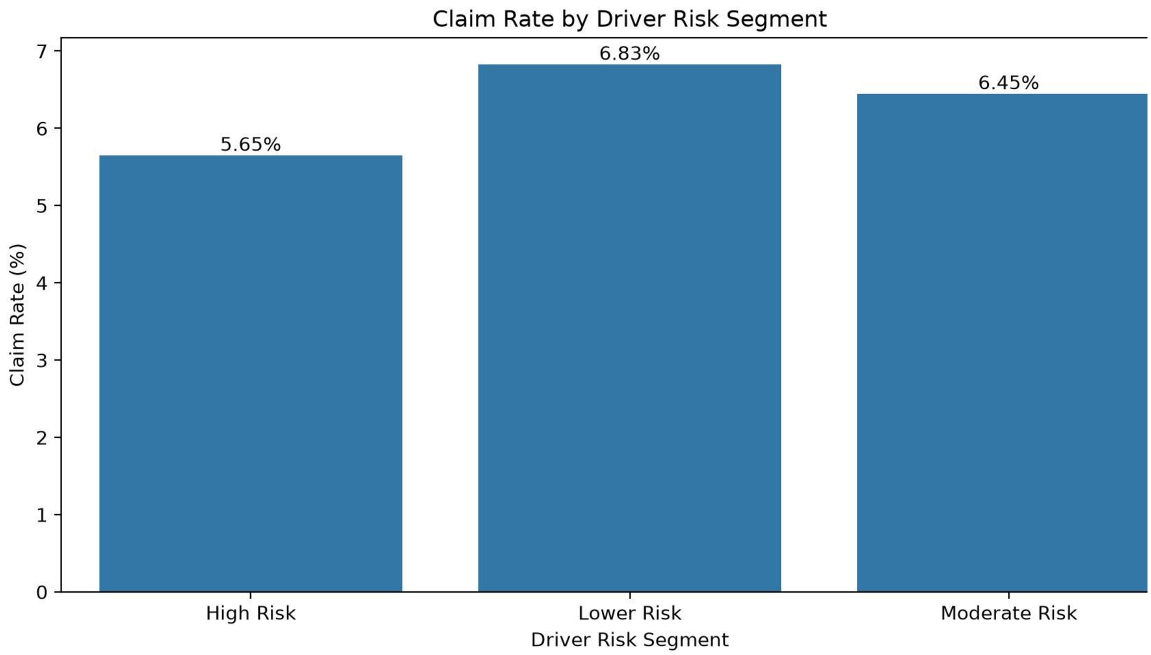
The portfolio contains 54,844 policies without a claim and 3,748 policies with a claim. The resulting 6.40% claim rate confirms a materially imbalanced target.



4.2 Driver Risk Segmentation

Observed claim rates are 5.65% for High Risk, 6.83% for Lower Risk and 6.45% for Moderate Risk. The non-monotonic pattern is important: the descriptive driver-risk label does not, by itself, provide a reliable ordering of observed claim incidence.

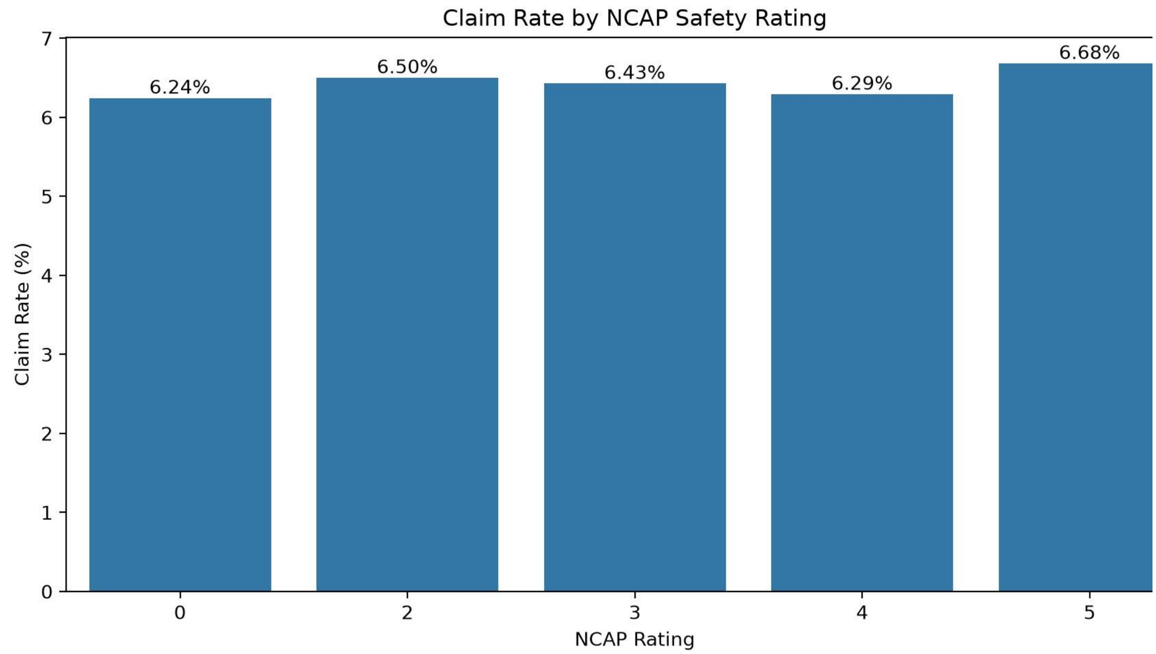
Driver risk segment	Policies	Observed claim rate
High Risk	10,038	5.65%
Lower Risk	13,433	6.83%
Moderate Risk	35,121	6.45%



4.3 NCAP Safety Rating

Observed claim rates range from 6.24% for NCAP rating 0 to 6.68% for rating 5. The relatively narrow spread suggests that NCAP rating should be treated as one input into a broader risk assessment rather than as a standalone claim rule.

NCAP rating	Policies	Observed claim rate
0	19,097	6.24%
2	21,402	6.50%
3	14,018	6.43%
4	2,114	6.29%
5	1,961	6.68%



4.4 Speeding Alert Analysis

The observed claim rate is 4.13% for speed-alert=0 and 6.41% for speed-alert=1. However, the alert=0 group contains only 363 policies compared with 58,229 alert=1 policies. The difference should therefore be treated as an observational signal requiring further validation, not as proof of causality.

4.5 Feature Engineering

- High_Risk_Driver_Flag was created as a combined risk-related feature.
- safety_equipment_count summarised relevant safety-equipment indicators.
- power_to_weight captured vehicle performance relative to vehicle weight.
- Additional transformations and feature-selection diagnostics were used to reduce redundancy and improve model readiness.

5. Model Results & Final Model Selection

Model	Test ROC-AUC	Precision	Recall	F1
Logistic Regression	0.586	0.080	0.547	0.140
Tuned Decision Tree	0.647	0.089	0.652	0.157

The Tuned Decision Tree was selected because it provides stronger unseen-data discrimination and claim capture than the Logistic Regression benchmark. Its ROC-AUC improves from 0.586 to 0.647 and recall from 0.547 to 0.652, while F1 improves from 0.140 to 0.157.

The precision of 0.089 is the key operational limitation. In practical terms, the model will generate many false alerts when attempting to capture a larger proportion of claims. Therefore, the appropriate business use is prioritisation and decision support, with human review for consequential underwriting actions.

5.1 Metric Reconciliation Note

A prediction-file-derived evaluation previously produced Precision 0.088, Recall 0.731 and F1 0.157. Because the 0.731 recall does not agree with the authoritative saved Stage 3 evaluation output, it has been excluded from the final business-report metrics. The final report consistently uses Precision 0.089, Recall 0.652, F1 0.157 and ROC-AUC 0.647. This prevents conflicting performance claims in the submission.

6. Dashboard & Management View

The Stage 4 dashboard should bring together portfolio KPIs, descriptive risk analysis and the model's predicted probability/risk tier. Management should be able to move from portfolio-level monitoring to segment-level analysis and then to policy-level underwriting review.

Dashboard view	Management purpose
Claim Distribution	Shows portfolio claim burden and class imbalance.
Driver Risk Segmentation	Compares observed claim rates across driver-risk labels.
NCAP Rating Analysis	Shows claim-rate variation across vehicle safety ratings.
Speeding Alert Analysis	Highlights the observed association requiring further validation.
Predicted Risk Level	Prioritises policies for review using the final model.
Probability Score	Shows the strength of the model's risk signal rather than only a class label.

7. Business Implications & Recommendations

7.1 Underwriting

- Use model probability and risk tier to prioritise manual underwriting review.
- Combine model output with policy, vehicle and safety information before material decisions.
- Do not automatically reject or accept a policy solely from the model prediction.

7.2 Pricing & Portfolio Management

- Use predicted risk as one input to actuarial segmentation, subject to regulatory and fairness validation.
- Monitor realised claim rates by predicted-risk tier before implementing differentiated pricing.
- Do not assume that descriptive labels such as High Risk necessarily imply higher claim incidence; the supplied EDA is non-monotonic.

7.3 Risk Prevention

- Use safety-related indicators to identify opportunities for targeted preventive engagement.
- Test whether interventions reduce subsequent claim frequency before scaling them.
- Investigate the speed-alert relationship using a larger and more balanced longitudinal sample.

7.4 Model Governance

- Monitor ROC-AUC, recall, precision, F1 and probability calibration after deployment.
- Monitor data drift and segment-level performance over time.
- Reassess thresholds against the financial cost of missed claims, false alerts and manual review.
- Perform fairness and subgroup-stability checks before production use.

8. Limitations & Next Steps

- The model has moderate discrimination (ROC-AUC 0.647), so it should not be presented as a highly accurate standalone claims predictor.
- Low precision means a substantial proportion of alerts may not become claims.
- Descriptive relationships do not establish causality.
- The driver-risk segmentation is non-monotonic with observed claims and should not be treated as a validated risk ranking.
- NCAP differences are relatively small in the supplied data.
- The speed-alert groups are highly imbalanced, requiring additional validation.
- Probability calibration, threshold optimisation using a business cost matrix, temporal validation and fairness testing are recommended next steps.
- The held-out prediction file should be regenerated from the exact Stage 3 final model/evaluation pipeline if a standalone prediction-output artifact is required for submission.

9. Conclusion

The analysis demonstrates a practical foundation for insurance claim-risk prioritisation. The Tuned Decision Tree is the preferred final model based on the authoritative held-out test evaluation. The model should be deployed as a decision-support layer that helps management prioritise underwriting attention, not as an autonomous decision engine. The dashboard and business recommendations provide the required bridge from model output to practical portfolio management.

Appendix — Final Authoritative Model Metrics

Metric	Logistic Regression	Tuned Decision Tree
Accuracy	0.569	0.551
Precision	0.080	0.089
Recall	0.547	0.652
F1	0.140	0.157
ROC-AUC	0.586	0.647

These values are the final figures to use consistently across the business report, dashboard narrative, presentation and submission documentation. The un-reconciled 0.731 recall must not be used as the final model metric.